

Problems 1 - 4

Week 38

Problem 1

Problem 1

Given $S = \{1,2,3,4,5\}$ and subsets

- $S_1 = \{1,2,5\}, w_1 = 2$
- $S_2 = \{1,3,4\}, w_2 = 4$
- $S_3 = \{2,5\}, w_3 = 5$
- $S_4 = \{2,4,5\}, w_4 = 6$
- $S_5 = \{3,4,5\}, w_5 = 8$
- $S_6 = \{4,5\}, w_6 = 3$

Use the Greedy algorithm to solve the set cover problem given. Slide 40 – 42, Chapter 1.

Solution

$$S_1 = \{1,2,5\}, w_1 = 2$$

$$S_2 = \{1,3,4\}, w_2 = 4$$

$$S_3 = \{2,5\}, w_3 = 5$$

$$S_4 = \{2,4,5\}, w_4 = 6$$

$$S_5 = \{3,4,5\}, w_5 = 8$$

$$S_6 = \{4,5\}, w_6 = 3$$

$$I \leftarrow \emptyset$$

$$\hat{S}_j \leftarrow S_j \quad \forall j$$

while I is not a set cover **do**

$$\ell = \arg \min_{j: \hat{S}_j \neq \emptyset} \frac{w_j}{|\hat{S}_j|}$$

$$I \leftarrow I \cup \{\ell\}$$

$$\hat{S}_j \leftarrow \hat{S}_j - S_\ell \quad \forall j$$

Problem 2

Problem 2

Given the following LP problem

$$\text{maximize } 4x_1 + 7x_2 + 6x_3$$

subject to

$$x_1 + 2x_2 + x_3 \leq 7$$

$$2x_1 + x_2 + 3x_3 \leq 14$$

$$x_1 + 4x_2 + 2x_3 \leq 11$$

$$x_1, x_2, x_3 \geq 0$$

- a) What is the objective function (objektfunkson)?
- b) What are the decision variables (beslutningsvariabler / avgjerdsvariablar)?
- c) What are the constraints (betingelser/begrensninger)?
- d) Give two examples of infeasible solutions (ulovlege/umulige løysingar).
- e) Give two examples of feasible solutions (lovlege/mulige).

Solution

Problem 3

Problem 3

Given $S = \{1,2,3,4,5\}$ and subsets

- $S_1 = \{1,2,5\}, w_1 = 2$
- $S_2 = \{1,3,4\}, w_2 = 4$
- $S_3 = \{2,5\}, w_3 = 5$
- $S_4 = \{2,4,5\}, w_4 = 6$
- $S_5 = \{3,4,5\}, w_5 = 8$
- $S_6 = \{4,5\}, w_6 = 3$

- a) Formulate the set cover problem as an integer linear program (ILP) given the sets above.
- b) Formulate the LP relaxation of the problem in a)

Solution

$$S_1 = \{1,2,5\}, w_1 = 2$$

$$S_2 = \{1,3,4\}, w_2 = 4$$

$$S_3 = \{2,5\}, w_3 = 5$$

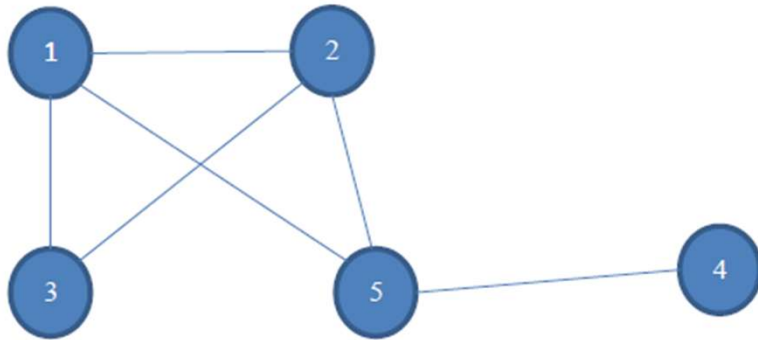
$$S_4 = \{2,4,5\}, w_4 = 6$$

$$S_5 = \{3,4,5\}, w_5 = 8$$

$$S_6 = \{4,5\}, w_6 = 3$$

Problem 4

A vertex cover of a graph is a subset of the vertices, $C \subseteq V$ such that every edge has at least one endpoint in C , that is $\forall (i,j) \in E$, either $i \in C$ or $j \in C$. We want to find a minimum vertex cover.



- a) Formulate the problem above as an integer linear program (ILP).
- b) Formulate the LP relaxation of the ILP above.

Solution

